



CampusFix

Smart Campus Complaint & Maintenance Tracker

Project Report

Course: Software Engineering

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Abstract

CampusFix is a desktop-based campus complaint and maintenance tracker created for a Software Engineering course project. The system focuses on one common university problem: students and staff often report maintenance issues informally, and those issues can be forgotten because there is no simple tracking flow. CampusFix provides a small but complete GUI application where users can submit complaints and administrators can assign, prioritize and close them. The project is implemented in Python using Tkinter for the graphical interface and SQLite for persistent local storage.

1. Introduction

In a campus environment, complaints about electricity, plumbing, classroom furniture, IT equipment, cleanliness and safety need a proper recording process. The aim of this project is not to build a large enterprise helpdesk, but to create a practical academic system that demonstrates requirements engineering, database design, GUI development, testing and documentation.

1.1 Project Objectives

- Provide a simple complaint submission process for students and staff.
- Allow administrators to view, filter, assign and update complaints.
- Maintain complaint history with timestamps and current status.
- Show dashboard statistics that support quick decision making.
- Keep the design moderate enough to explain in a class presentation and viva.

1.2 Team Details

Member	Roll Number	Main Responsibility
Muhammad Azam	IU05-0124-0190	Database design, testing support and documentation review
Kashif Raza	IU05-0124-0303	GUI design, requirements analysis and final integration
Tehreem Ijaz	IU05-0124-0126	SRS preparation, use cases and report formatting

2. Problem Statement

Manual complaint handling creates delays because complaints may be shared verbally, through WhatsApp messages or on paper. These methods do not provide a clear record of who submitted the complaint, where the issue is located, what priority it has, and whether the maintenance team has resolved it. CampusFix solves this by turning the complaint process into a small structured workflow.

3. Proposed Solution

The proposed system is a Python desktop application with two main roles: User and Admin. A user can register, log in, submit a complaint and view personal complaints. An admin can log in, see all complaints, search/filter records and update complaint status, assigned person and notes. SQLite is used as the local database because it is lightweight and easy to demonstrate on any laptop.

4. Technology Stack

Layer	Technology	Reason
Programming Language	Python	Readable syntax and suitable for quick GUI/database development.
GUI Framework	Tkinter	Built into Python, so the project runs without heavy external setup.
Database	SQLite	Lightweight file-based database, good for academic desktop applications.
Testing	unittest	Used to validate main database operations without extra dependencies.
Architecture	Layered desktop architecture	Separates interface code from database operations for maintainability.

5. SDLC Model Used

The project follows an Iterative and Incremental Model. In this model, the system is developed in small working parts instead of waiting until the end to test everything. CampusFix was divided into increments such as database setup, authentication, complaint submission, admin dashboard, search/filtering and testing.

5.1 Why This Model Was Used

- The project requirements were understandable, but the GUI needed small improvements after testing each screen.
- It allowed the team to create a working version early and then improve it step by step.
- Each module could be tested separately, especially database functions and role-based workflows.
- It fits a semester project because progress can be shown in milestones rather than only at the final stage.

6. Scope

The scope includes registration, login, complaint creation, complaint listing, admin filtering, status updates, assignment, notes and dashboard statistics. The current version is designed for a single campus and local desktop use. Online hosting, email notifications, mobile application support and file attachments are outside the current scope and are listed as future enhancements.

7. Functional Requirements

ID	Requirement	Description
FR-01	User Registration	The system shall allow new student/staff users to create an account with name, email and password.
FR-02	Login	The system shall authenticate users and open the correct dashboard based on role.
FR-03	Submit Complaint	A user shall submit a complaint with title, category, location, priority and description.
FR-04	Track Complaints	A user shall view his or her complaint history and current status.
FR-05	Admin View	An admin shall view all submitted complaints in one dashboard.
FR-06	Search and Filter	An admin shall filter complaints by status and search by title, reporter, category or complaint ID.
FR-07	Assign Complaint	An admin shall assign a complaint to a staff member or technician.
FR-08	Update Status	An admin shall update the status as Submitted, Assigned, In Progress, Resolved or Closed.
FR-09	Dashboard Counts	The system shall display total, open, resolved and urgent complaint counts.

8. Non-Functional Requirements

ID	Attribute	Description
NFR-01	Usability	The interface should be simple enough for a first-time user to submit a complaint without training.
NFR-02	Reliability	SQLite foreign keys and required fields should protect the basic consistency of complaint records.
NFR-03	Performance	Common dashboard operations should respond quickly for a small campus dataset.
NFR-04	Maintainability	The code should keep GUI, configuration and database functions separate.
NFR-05	Security	Passwords are not stored directly; the project stores SHA-256 password hashes for demonstration.

9. System Diagrams

9.1 Use Case Diagram

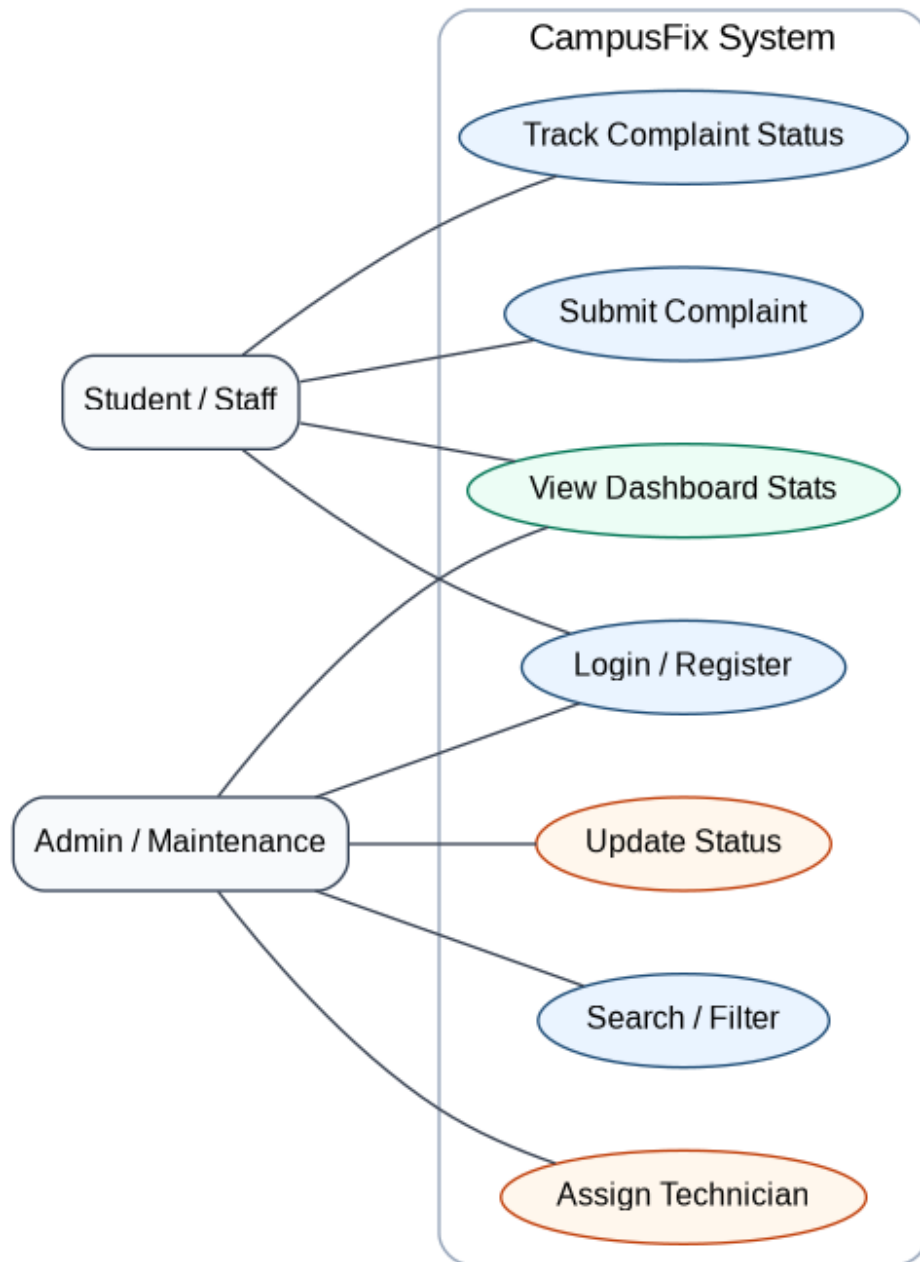


Figure 1: Main actors and use cases of CampusFix.

9.2 Class Diagram

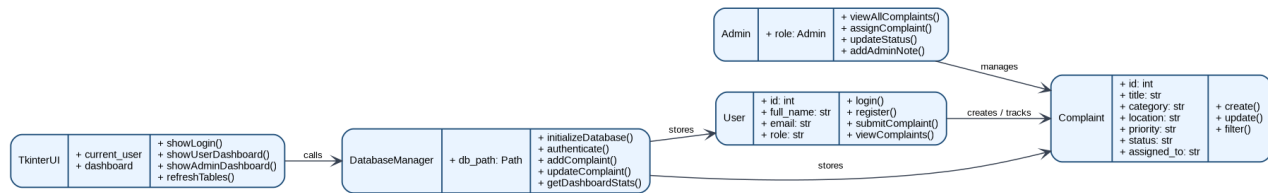


Figure 2: Main classes/modules and their responsibilities.

9.3 Activity Diagram

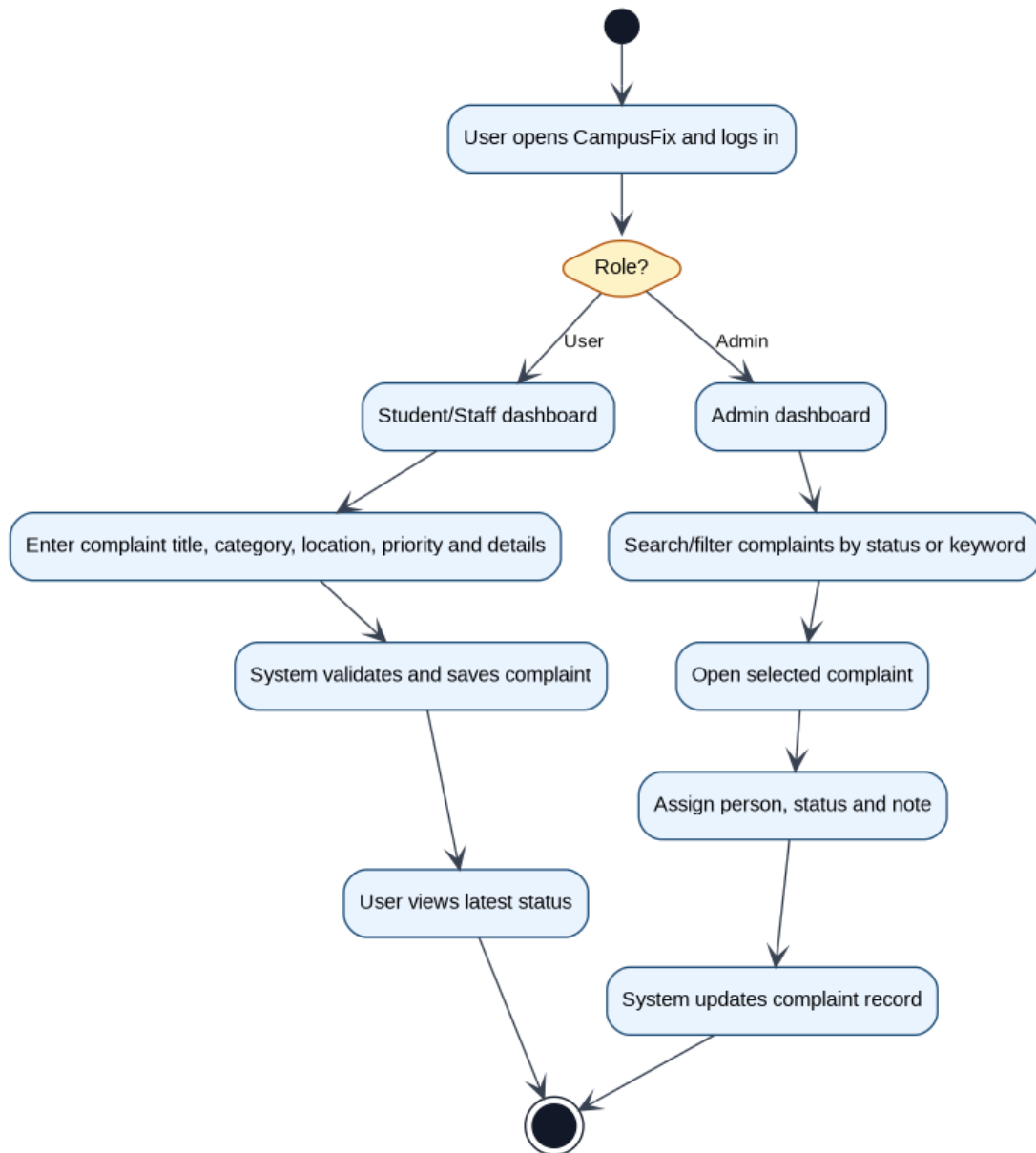


Figure 3: Complaint workflow for user and admin paths.

9.4 Sequence Diagram

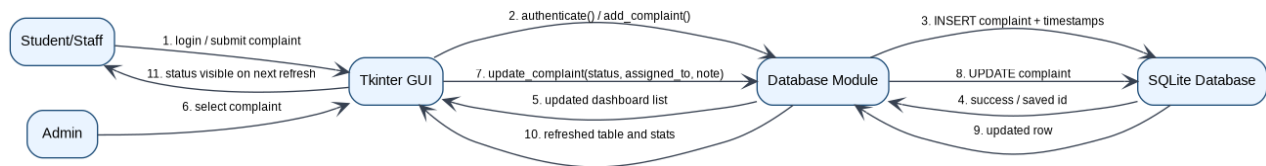


Figure 4: Sequence of complaint submission and admin update.

9.5 UML Component Diagram

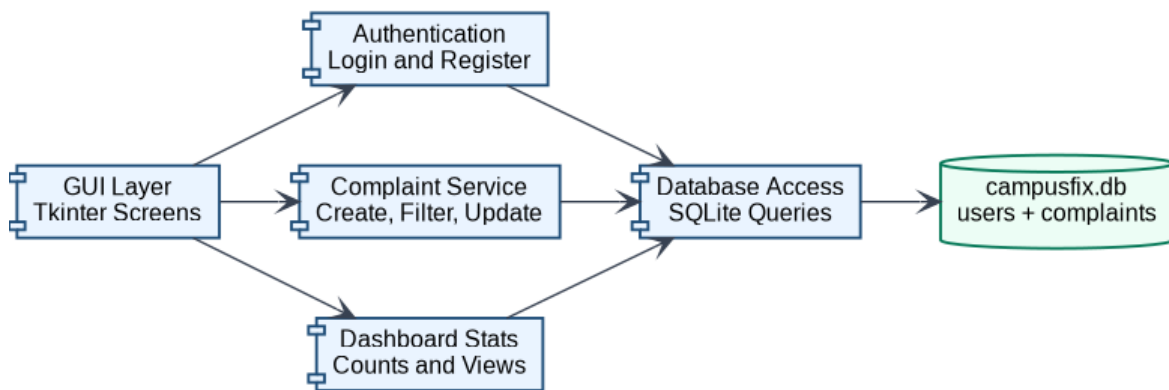


Figure 5: UML component view showing GUI, services and database layer.

9.6 Entity Relationship Diagram (ERD)

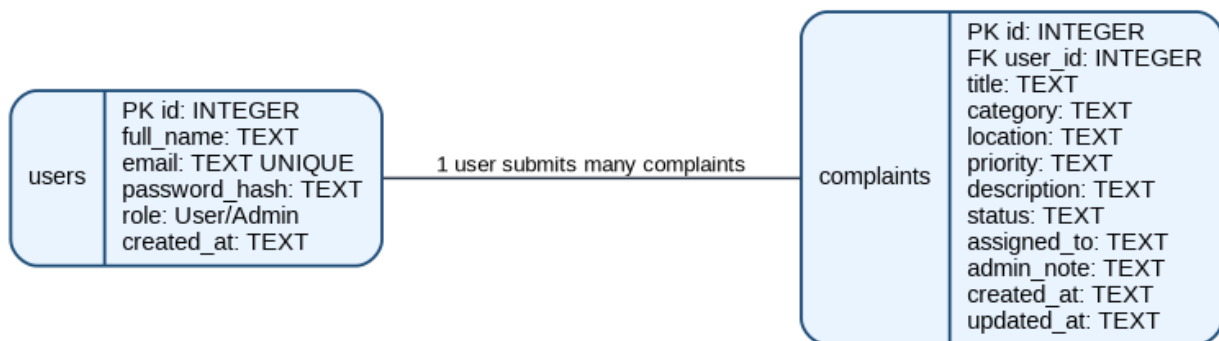


Figure 6: Database relationship between users and complaints.

10. Database Design

Table	Important Fields	Purpose
users	id, full_name, email, password_hash, role, created_at	Stores login accounts and user roles.
complaints	id, user_id, title, category, location, priority, description, status, assigned_to, admin_note, created_at, updated_at	Stores complaint records and admin progress updates.

The complaints table has a foreign key user_id that connects each complaint with the user who submitted it. This design keeps the database simple while still supporting the most important reporting and tracking operations.

11. System Features

Login and Registration

The system supports role-based login. The default admin and demo student accounts make classroom demonstration easier.

User Dashboard

Users can add complaints and view complaint status in a table with latest timestamps.

Admin Dashboard

Admins can search, filter, assign and update complaints without directly editing the database.

Dashboard Statistics

The application shows total, open, resolved and urgent counts to make complaint load visible.

Local Persistence

All records are stored in campusfix.db, so the application retains data after closing.

12. User Interface Screens

CampusFix • Smart Campus Complaint & Maintenance Tracker
Logout

Signed in as Demo Student (User)

Total Requests
2

Open
2

Resolved
0

Urgent Open
0

+ New Complaint

Refresh

My Complaints

ID	Title	Category	Priority	Status	Assigned To	Created
2	WiFi weak in library	IT / Network	Medium	Submitted	-	2026-09-04 15:33:56
1	Projector not working	Classroom Equipment	High	Submitted	-	2026-09-04 15:33:56

Figure 7: User dashboard for submitting and tracking complaints.

CampusFix • Smart Campus Complaint & Maintenance Tracker
Logout

Signed in as Campus Administrator (Admin)

All Requests
2

Open
2

Resolved
0

Urgent Open
0

Status: All Search: Apply Clear

Complaint Queue

ID	Title	Reporter	Category	Priority	Status	Assigned To	Created
1	Projector not working	Demo Student	Classroom Equipment	High	Submitted	-	2026-09-04 15:33:56
2	WiFi weak in library	Demo Student	IT / Network	Medium	Submitted	-	2026-09-04 15:33:56

Open / Update Selected

Refresh

Figure 8: Admin dashboard for filtering and updating complaint records.

13. Testing

Test Case	Input / Action	Expected Result	Status
TC-01	Login with valid default admin account	Admin dashboard opens	Pass
TC-02	Register a new user with unique email	User account is created	Pass
TC-03	Submit a complaint with required values	Complaint ID is created and saved	Pass
TC-04	Filter complaints by status	Only matching complaints are shown	Pass
TC-05	Update status and assigned person	Complaint record shows new status and note	Pass
TC-06	Dashboard count after complaint update	Statistics refresh correctly	Pass

Automated database tests were also included to check registration, authentication, complaint creation, status update and dashboard statistics. This makes the project easier to verify during development.

14. Limitations

- The current version is a desktop application, not a web-hosted system.
- No email or SMS notification module is implemented.
- No image attachment feature is included for complaint evidence.
- The admin role is seeded for demonstration rather than managed through a full user management screen.

15. Future Enhancements

- Add photo attachments for complaints.
- Add email notifications when status changes.
- Add a web dashboard using Flask or Django.
- Add monthly reports for complaint categories and response time.
- Add technician login so assigned staff can update their own tasks.

16. Conclusion

CampusFix is a compact but complete Software Engineering project. It shows how a real campus problem can be converted into requirements, diagrams, database tables, a working GUI and test cases. The system is intentionally not overcomplicated, which makes it suitable for explanation in a class presentation and viva.